INDEPENDENT PROJECT · PUBLIC BENCHMARK DATA

Industrial Visual Anomaly Detection

From Requirements to a Tested Prototype

Flags printed-circuit-board images for manual review. Built on the public VisA benchmark (Amazon Science, CC BY 4.0) with AI coding assistance; no factory data was used.

Executive summary

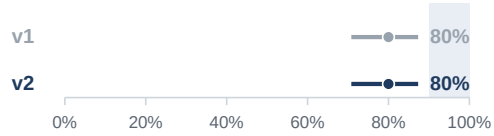
Developed and evaluated an AI-assisted visual inspection prototype using public printed-circuit-board images. A targeted refinement reduced false alarms from 13% to 3% on a board type that was held back and evaluated independently, while defect recall remained 80%. The prototype is below the proposed 90% recall target and remains sensitive to darker images. Further improvement is required before a manufacturing pilot.

Results at a glance

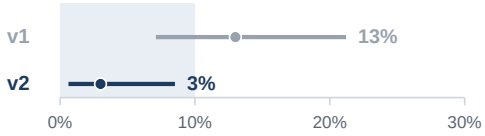
Board type PCB2, official test set: 100 defective and 100 normal images. v1 = first version, v2 = refined version; review thresholds fixed before testing.

Defects flagged (recall)	False-alarm rate	Ranking quality (AUROC)	Median time per image
80% → **80%**	13% → **3%**	0.927 → **0.979**	194 → **224**
of 100 defective images	of 100 normal images	1.0 = perfect separation	milliseconds, one GPU
target 90%: **Not met**	target 10% or less: **Met**	target 0.90: **Met**	target 1 s or less: **Met**

Defects flagged for review (recall)



False alarms, official-test normal images



Shaded: proposed target. Lines: 95% confidence ranges (80% recall on 100 defective images is compatible with roughly 71% to 87%). A separate check on 151 further normal images, outside the official test, gave false-alarm rates of 5.3% (v1) and 4.0% (v2).

What changed and why

The first version sometimes treated background texture, such as dust on the cloth under the board, as a defect. The second version focuses anomaly scoring on the board region, reducing false alarms without improving missed-defect detection; most remaining misses are very small defects. The change was fixed in advance and tested once on a held-back board type.

Reliability and go / no-go

Recall ≥ 90%	**Not met** · 80% of defects flagged
False alarms ≤ 10%	**Met** · 3% official test; 4.0% separate check
Lighting robustness	**Not met** · 10% darker: false alarms 4% → 29% (251 normal images)

Formal evaluation (rules fixed before testing): the expected recall gain was not demonstrated. Rule-based verdict: “Inconclusive”. Targets were missed, but no automatic stop rule fired, because recall sits exactly at the 80% minimum; this does not mean the criteria were met.

> **Recommendation: not ready for a manufacturing pilot.** Although v2 improved false-positive performance, it did not meet the proposed detection-sensitivity and robustness requirements. It is therefore not recommended for a manufacturing pilot without further work.

Engineering readiness

Reproducible evaluation: every number on this page is regenerated from saved per-image results and checked automatically. **Frozen thresholds** were recorded before any test image was scored. **Automated tests**, **documented assumptions** and **engineering handoff** notes are part of the repository.

Limitations and next steps. Two board types from one public benchmark, laboratory images, 100 defective images per board type; no transfer to other products, cameras or factories is implied. Next: lighting normalisation, better small-defect sensitivity, and targets confirmed by a real manufacturing user, each followed by a fresh, independent evaluation.

Data: VisA (Amazon Science), CC BY 4.0. Pretrained network: torchvision Wide-ResNet-50-2 (ImageNet weights). Methods, statistics and reproduction: project repository (README, docs/v2_decision_record.md). No endorsement by the dataset authors or any company is implied.
Project period: October 2026. Delivery time: pending owner review.